

Thermal Hydraulics for Reactor Kinetics Lab

1 Purpose

This note is a first-pass thermal-hydraulics design reference for the *Reactor Kinetics Lab* repository. Its purpose is not to produce a final plant design. Instead, it does three narrower things:

1. explains why the current Modelica prototype overheats so quickly,
2. derives rough coolant-capacity targets from the repository's existing 20 MW annular-core assumptions,
3. converts those targets into a more realistic Modelica subsystem structure for the next implementation pass.

The intended use is practical: this note should give enough engineering scale to replace the current toy loop with a research-reactor-like thermal-hydraulic model that stays within plausible operating temperatures.

2 Current mismatch

The current Python neutronics model assumes a reactor with:

- nominal thermal power of 20 MW,
- annular active core geometry with $R_{\text{inner}} = 0.8$ m, $R_{\text{fuel}} = 3.456$ m, and active height $H = 6.912$ m,
- heavy-water moderation as the frozen physics assumption.

The current Modelica primary-loop prototype is useful as a numerical proof of concept, but it is still a small pedagogical loop. In particular, the present prototype:

- uses a single equivalent heated pipe rather than a reactor-scale coolant channel network,
- uses a simple radiator-like sink rather than a reactor heat exchanger,
- has no explicit saturation-margin target,
- does not yet enforce a reactor-appropriate pressure regime.

That is why the hot-leg temperature can shoot toward values such as 400°C even while the return side remains much cooler. The prototype is absorbing heat numerically, but it is not yet constrained by the real coolant inventory, realistic flow capacity, or a physically chosen primary operating envelope. A real low-pressure research reactor would not tolerate a bulk outlet temperature anywhere near this regime; long before that point, boiling, voiding, structural damage, and severe operational upset would occur.

3 Reactor-scale inputs

For this note, the frozen reactor-scale inputs are taken directly from the existing repository documentation:

Quantity	Value
Nominal thermal power P	20 MW
Inner moderator radius R_{inner}	0.8 m
Fuel outer radius R_{fuel}	3.456 m
Active height H	6.912 m
Primary coolant assumption	D ₂ O

The active annular volume is approximated by

$$V_{\text{core}} = \pi \left(R_{\text{fuel}}^2 - R_{\text{inner}}^2 \right) H.$$

Using the current geometry,

$$V_{\text{core}} \approx \pi \left(3.456^2 - 0.8^2 \right) \times 6.912 \approx 245.5 \text{ m}^3.$$

This gives an average volumetric power density of

$$q''' = \frac{P}{V_{\text{core}}} \approx \frac{20 \times 10^6}{245.5} \approx 8.15 \times 10^4 \text{ W/m}^3 = 0.0815 \text{ MW/m}^3.$$

This value is not extreme for a large, heavily moderated core. The main issue is therefore not that the core is too power-dense, but that the present prototype loop is not sized to move reactor-scale heat with reactor-scale coolant flow.

4 Cooling-capacity estimates

4.1 Bulk coolant temperature rise

For a first-pass heavy-water estimate, it is acceptable to use water-like bulk properties for sizing:

$$c_p \approx 4.2 \text{ kJ/(kg K)}, \quad \rho \approx 1100 \text{ kg/m}^3.$$

The required mass flow for a chosen bulk temperature rise is approximated by

$$\dot{m} = \frac{Q}{c_p \Delta T}.$$

For $Q = 20 \text{ MW}$, the required total primary-loop flow is:

Bulk rise ΔT	$\dot{m}$ [kg/s]	$\dot{V}$ [m ³ /s]
10 K	476.2	0.4329
20 K	238.1	0.2165
30 K	158.7	0.1443
40 K	119.0	0.1082
50 K	95.2	0.0866

The most useful design range for the next Modelica pass is the middle of this table, not the extremes. A 20 K to 30 K bulk rise is a reasonable first target for a research-reactor-like forced-circulation system. That implies a total primary mass flow on the order of 160 to 240 kg/s.

4.2 Recommended bulk temperature envelope

Because the current repository concept is much closer to a research reactor than to a high-temperature power reactor, the next Modelica model should target a bulk operating envelope well below the saturation concerns implied by a 400°C hot leg. A practical first-pass envelope is:

Quantity	Suggested target
Primary inlet bulk temperature	45–60°C
Primary outlet bulk temperature	65–90°C
Bulk rise across core	20–30 K
Pressure regime	low-pressure or modestly pressurized

This does not prove safety margins, but it does set a sane target for the next Modelica architecture: the hot leg should stay in a low-temperature, single-phase, research-reactor regime rather than drifting toward a PWR-like high-temperature regime that the current design does not support.

4.3 Heat-transfer surface argument

If the annulus were naively cooled only through its inner and outer cylindrical boundaries, the corresponding bounding area would be

$$A_{\text{bounds}} = 2\pi(R_{\text{inner}} + R_{\text{fuel}})H \approx 184.8 \text{ m}^2.$$

At 20 MW, that would correspond to an average heat flux of

$$q'' \approx \frac{20 \times 10^6}{184.8} \approx 108 \text{ kW/m}^2.$$

This is already enough to show that the next Modelica model should not be built as one tiny equivalent pipe with one tiny equivalent wall area. A real core would expose a much larger wetted area through many coolant channels, plates, or subassemblies. Therefore the next thermal-hydraulic abstraction should move toward *parallel channels plus plena*, not a single narrow toy conduit.

4.4 Pump-duty scale

The hydraulic power scale is also useful for planning. Using

$$W_{\text{pump}} \approx \frac{\Delta p \dot{V}}{\eta},$$

with $\eta \approx 0.7$ and the flow rates above, the shaft-power scale is on the order of:

Bulk rise ΔT	$\Delta p = 0.5 \text{ MPa}$	1.0 MPa	2.0 MPa
20 K	155 kW	309 kW	618 kW
30 K	103 kW	206 kW	412 kW
40 K	77 kW	155 kW	309 kW

These are only order-of-magnitude values, but they are already enough to guide the next Modelica system away from a tiny pump and toward a reactor-scale head curve with meaningful coastdown behavior.

5 Candidate primary-loop architectures

5.1 Option A: low-pressure research-reactor concept

The first candidate is a low-pressure, large-inventory heavy-water system with forced circulation through a channelized core and a separate heat sink. This is the option that best matches the current repository’s research-reactor framing.

Its main features would be:

- inlet and outlet plena with significant coolant inventory,
- many effective coolant channels through the core,
- a forced-circulation pump train,
- a primary heat exchanger or cooler outside the core,
- bulk temperatures kept comfortably below boiling.

This option is the most natural follow-on to the existing open-tank intuition, but it must become much more structured than the current prototype. In particular, the “open tank + radiator” picture should evolve into “plenum/vessel volumes + channel bank + dedicated heat sink.”

5.2 Option B: closed forced-circulation loop with secondary side

The second candidate is a more explicitly closed D₂O primary loop feeding a secondary heat exchanger. This offers cleaner control over pressure and heat rejection, and it is easier to connect to a later FMU co-simulation because the boundaries are more explicit.

Its main features would be:

- a closed primary inventory,
- one or more pumps with a realistic head curve,
- explicit primary-side heat exchanger,
- secondary-side sink represented either by a fixed-temperature sink or a simplified secondary loop.

This option is more structured and may eventually be better for FMI coupling, but it also moves the model farther from the very simple research-reactor-like mental model the current prototype was trying to preserve.

6 Recommended operating envelope and structure

For the next Modelica implementation, the recommended direction is:

Use a **low-temperature forced-circulation D₂O primary model** with **channelized core flow, plena or vessel volumes**, and an **explicit heat sink**, while keeping bulk outlet temperatures in the roughly 65–90°C range rather than allowing a high-temperature pressurized-loop regime to emerge by accident.

That recommendation is based on three observations:

1. the reactor-scale power density is modest enough that a large-flow, low-temperature solution is plausible,
2. the current repository already assumes a research reactor, not a compact power reactor,
3. the present Modelica overheating is better interpreted as a missing thermal-hydraulic capacity model than as evidence that the reactor must operate at very high primary temperatures.

7 Modelica implementation targets

The next thermal-hydraulic Modelica model should no longer be a single heated pipe plus radiator. A better first abstraction is:

1. **inlet plenum or lower vessel volume**,
2. **equivalent channel bank** through the core, either as parallel channels or one channel with explicit scaling parameters,
3. **outlet plenum or upper vessel volume**,
4. **pump model** with a head curve and optional coastdown,
5. **primary heat exchanger or cooler** that removes the full 20 MW,
6. **pressure boundary strategy** consistent with a low-pressure or modestly pressurized research-reactor system.

The minimum FMI-oriented variable set should be:

Direction	Variables
Python $\rightarrow$ Modelica	total core power, axial power fractions, control actions
Modelica $\rightarrow$ Python	inlet temperature, outlet temperature, total flow, pressure drop

For the channel abstraction, the following scaling logic is recommended:

- total loop flow should be a reactor-scale quantity, about 160–240 kg/s for a 20–30 K rise,
- core heating should be distributed axially, as already planned,
- the effective coolant area and wetted surface should represent many channels, not one narrow laboratory pipe.

8 Immediate implications for the current prototype

The present Modelica prototype is still useful, but it should now be interpreted as a *numerical topology prototype*, not as a thermal-hydraulic design. Before adding full neutronics feedback, the next Modelica pass should at least:

1. scale the nominal power from 10 kW-like behavior to the full 20 MW,
2. replace the tiny equivalent core pipe with a channel-bank abstraction,
3. replace the radiator sink with a reactor-scale heat sink or exchanger,
4. enforce a low-temperature operating envelope that prevents unrealistic hot-leg temperatures.

9 Open questions

This note deliberately leaves several questions open for the next iteration:

- Should the next primary model be low-pressure and open/pool-like, or slightly pressurized but still firmly in a research-reactor regime?
- How many effective parallel coolant channels are needed to make the channel-bank abstraction believable without introducing unnecessary numerical stiffness?
- Is a heavy-water property package needed immediately, or is water-like property approximation acceptable for the first reactor-scale thermal sizing pass?

The key point is that these are now engineering-structure questions, not a sign that the reactor should be allowed to run at several hundred degrees Celsius in the primary loop.